

Affine-DDPNM: Methodology

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We extend the domain-decomposition pore-network method (DD-PNM) of [1] by enriching the interface traction space from a single constant-normal mode to a nine-mode affine representation.

1 Model problem & sign convention

Steady incompressible Stokes flow in $\Omega \subset \mathbb{R}^3$:

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, \quad \text{in } \Omega, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega, \quad (2)$$

$$\mathbf{u} = \mathbf{0}, \quad \text{on } \Gamma_{\text{wall}}, \quad (3)$$

$$-\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = p_b \mathbf{n}, \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (4)$$

$$(I - \mathbf{n} \otimes \mathbf{n}) \boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = \mathbf{0}, \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (5)$$

where $\boldsymbol{\sigma}(\mathbf{u}, p) := -p\mathbf{I} + \nu \nabla \mathbf{u}$ is the pseudostress and $p_b = p_{\text{in}}$ on Γ_{in} , $p_b = p_{\text{out}}$ on Γ_{out} . The inlet/outlet condition **prescribes the normal pseudotraction** $-\sigma n = p_b n$, not the pressure directly; the distinction matters when $\nu \nabla \mathbf{u}$ contributes to the normal stress.

2 Domain partition

Ω is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$. Interface $\gamma_k = \partial\Omega_i \cap \partial\Omega_j$ ($k = 1, \dots, m$) is assumed **planar** throughout the theoretical development (see Remark 1 for watershed). Each Ω_i has m_i ports, m_i^u internal and m_i^k boundary. Boolean restriction maps R_i^u, R_i^k follow [1].

For the Voronoi partition, each interface is the perpendicular-bisector plane of the two sphere centres; for unequal radii this differs from the exact clearance saddle surface. The geometric error is quantified in the numerical companion.

3 Reference FE discretization

Taylor–Hood $[P_2]^3$ – P_1 on body-fitted tetrahedral meshes. Velocity space $\mathbf{V}_h$, pressure space $Q_h = P_1$. On Ω_i with Neumann data:

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_i \\ \mathbf{P}_i \end{bmatrix} = \begin{bmatrix} \sum_{r=1}^{m_i} \mathbf{f}_i^{(r)} \\ \mathbf{0} \end{bmatrix}. \quad (6)$$

No pressure stabilisation is used.

4 Interface traction Ansatz

4.1 Classic DD-PNM

$$-\sigma(\mathbf{u}, p) \mathbf{n}_\gamma = p_\gamma \mathbf{n}_\gamma, \quad (7)$$

one scalar p_γ per interface.

4.2 Affine-DDPNM (nine modes per interface)

For each planar interface γ , fix an orthonormal frame $\{\mathbf{n}_\gamma, \mathbf{t}_{\gamma,1}, \mathbf{t}_{\gamma,2}\}$ with $\mathbf{n}_\gamma$ pointing from the first to the second pore of the pair. Half-span-normalised coordinates:

$$s(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_\gamma) \cdot \mathbf{t}_{\gamma,1}}{a_\gamma}, \quad t(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_\gamma) \cdot \mathbf{t}_{\gamma,2}}{b_\gamma}, \quad (8)$$

with a_γ, b_γ the half-spans, guaranteeing $s, t = O(1)$ under shape-regularity.

Signed direction vectors for pore i :

$$\mathbf{d}_{i,\gamma,\mathbf{n}} = \mathbf{n}_{i,\gamma}, \quad \mathbf{d}_{i,\gamma,\mathbf{t}_\nu} = \sigma_{i,\gamma} \mathbf{t}_{\gamma,\nu} \quad (\nu = 1, 2), \quad (9)$$

where $\sigma_{i,\gamma} = +1$ (-1) if i is the first (second) member. These satisfy $\mathbf{d}_{j,\gamma,\beta} = -\mathbf{d}_{i,\gamma,\beta}$.

The **applied traction** on γ as seen from Ω_i is

$$-\sigma(\mathbf{u}, p) \mathbf{n}_{i,\gamma} = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{\gamma,\alpha,\beta} \varphi_\alpha(s, t) \mathbf{d}_{i,\gamma,\beta}, \quad (10)$$

where $\mathcal{M} = \{0, s, t\} \times \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$, $\varphi_0 = 1$, $\varphi_s = s$, $\varphi_t = t$. The minus sign matches the classic convention (7). Nine coefficients $\lambda_\gamma \in \mathbb{R}^9$ per interface.

Remark 1 (Planar interface requirement). *The theory assumes planar interfaces with fixed $\mathbf{n}_\gamma$. Watershed partitions produce bent interfaces where the mesh-level facet normal varies. In the code, watershed interfaces use per-facet $\mathbf{n}(x)$ for the normal load and a fixed averaged frame for tangent directions; this deviates from the nine-mode space and is one factor behind the degraded Affine accuracy on watershed geometries.*

5 Local compliance map

5.1 Primitive response loads

For mode (α, β) on internal interface γ of Ω_i :

$$[\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}]_j = \int_\gamma \varphi_\alpha(s, t) \mathbf{d}_{i,\gamma,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \, dS. \quad (11)$$

With the sign convention (10), a positive λ acts into the pore; the load vector carries no leading minus sign—it is the right-hand side of (6) directly. For boundary ports only $(\alpha, \beta) = (0, \mathbf{n})$ is used with a prescribed coefficient.

Solve for each unit load:

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{X}_{i,\gamma}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}. \quad (12)$$

5.2 Compliance map and weighted velocity moments

$n_i = 9m_i^u + m_i^k$ local coefficients (m_i^k of them prescribed). $L_i \in \mathbb{R}^{n_i, u \times n_i}$: columns = all $\mathbf{f}_i^{(\mu)}$.

Velocity: $\mathbf{U}_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i, \gamma, \alpha, \beta} \mathbf{U}_{i, \gamma}^{(\alpha, \beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i, \gamma}^k \mathbf{U}_{i, \gamma}^{(0, \mathbf{n})}$.

Weighted velocity moments on interface γ :

$$M_{i, \gamma}^{(\alpha, \beta)} = \int_{\gamma} \varphi_{\alpha}(s, t) \mathbf{u}_{i, h} \cdot \mathbf{d}_{i, \gamma, \beta} dS. \quad (13)$$

$M_{i, \gamma}^{(0, \mathbf{n})}$ = volumetric flow rate (the only quantity with the physical meaning of a flux). $M^{(s, \mathbf{n})}, M^{(t, \mathbf{n})}$ are linear normal flux moments; the six $M^{(\alpha, t_{\nu})}$ are tangential velocity moments. Collectively: **nine weighted velocity moments**.

The compliance (Neumann-to-Dirichlet) map:

$$\mathbf{G}_i = L_i^{\top} H_i L_i \in \mathbb{R}^{n_i \times n_i}, \quad H_i = A_i^{-1} - A_i^{-1} B_i^{\top} M_i^{-1} B_i A_i^{-1}. \quad (14)$$

$\mathbf{G}_i \succeq 0$, symmetric. Partitioned into $\mathbf{G}_i^{uu}, \mathbf{G}_i^{uk}$.

6 Global Schur complement assembly

Enforcing continuity of **all nine weighted velocity moments** ($M_{i, \gamma}^{(\alpha, \beta)} + M_{j, \gamma}^{(\alpha, \beta)} = 0$):

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^{\top} \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = - \sum_{i=1}^N (R_i^u)^{\top} \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k. \quad (15)$$

$\mathbf{S} \in \mathbb{R}^{9m \times 9m}$, symmetric by construction.

Remark 2 (Conditions for SPD). $\mathbf{S}$ is positive definite under the following conditions:

1. **Global anchoring:** every connected component of the pore adjacency graph contains at least one inlet/outlet port;
2. **Local coercivity:** every pore has a positive-measure no-slip wall segment;
3. **Non-degenerate loads:** $\text{rank}(H_i^{1/2} L_i) = n_i - 1$ for every pore (verified numerically across all tested geometries).

Conditions (1) and (2) are distinct: (2) ensures K_i is invertible; (1) eliminates the global constant-pressure kernel mode of $\mathbf{S}$. In the watershed partition, floating basins (pores lacking a solid-wall facet) violate condition (2) and are merged into neighbours before assembly. This restores local coercivity but does not supply condition (1); the inlet/outlet ports independently anchor the global system. **Per-face linear independence** of the nine primitive loads is necessary but not sufficient for condition (3), which additionally requires the absence of inter-interface linear dependencies after projection through $H_i^{1/2}$.

7 Solution algorithm

Offline library (per pore, in parallel)

- (a) Assemble A_i, B_i .
- (b) For each internal γ and $(\alpha, \beta) \in \mathcal{M}$: assemble $\mathbf{f}_{i, \gamma}^{(\alpha, \beta)}$, solve for $(\mathbf{U}, \mathbf{X})$.
- (c) For each boundary γ : solve $(0, \mathbf{n})$ load.
- (d) Form $\mathbf{G}_i$, accumulate $\mathbf{S}, \mathbf{F}$.

2. **Global solve** $\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}$ (sparse direct).

Reconstruction (in parallel)

$$\begin{aligned}\mathbf{U}_i &= \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i, \gamma, \alpha, \beta} \mathbf{U}_{i, \gamma}^{(\alpha, \beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i, \gamma}^k \mathbf{U}_{i, \gamma}^{(0, \mathbf{n})}, \\ \mathbf{P}_i &= \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i, \gamma, \alpha, \beta} \mathbf{X}_{i, \gamma}^{(\alpha, \beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i, \gamma}^k \mathbf{X}_{i, \gamma}^{(0, \mathbf{n})}.\end{aligned}$$

8 Relation to the classic DD-PNM

The classic traction space is a subspace of the affine space ($\mathcal{T}_\gamma^{\text{P0}} \subset \mathcal{T}_\gamma^{\text{Affine}}$). The classic DD-PNM is recovered by **restricting both trial and test spaces**:

$$\mathbf{S}_{\text{P0}} = \mathcal{P}_{\text{P0}}^\top \mathbf{S} \mathcal{P}_{\text{P0}}, \quad \mathbf{F}_{\text{P0}} = \mathcal{P}_{\text{P0}}^\top \mathbf{F},$$

where $\mathcal{P}_{\text{P0}}$ embeds the m classic coefficients into $\mathbb{R}^{9m}$ by placing them in the $(0, \mathbf{n})$ slots. The classic solution $\mathbf{p} = \mathbf{S}_{\text{P0}}^{-1} \mathbf{F}_{\text{P0}}$ generally differs from the $(0, \mathbf{n})$ -components of $\boldsymbol{\lambda} = \mathbf{S}^{-1} \mathbf{F}$: the eight additional test-space constraints alter the Galerkin projection. The enrichment guarantee is that $\text{Range}(\mathcal{P}_{\text{P0}}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$, so the affine error in the energy (semi-)norm is never larger than the classic error (see the companion mathematical properties document).

9 Computational cost

$n_i \approx 9m_i$ (vs. m_i classic). Offline: n_i independent local Stokes solves (embarrassingly parallel). Online: sparse LU of $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$, with at most $9 \times (\text{max coordination}) \times 9$ non-zero blocks per row. For multi-boundary-condition sweeps the offline library is built once.

References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.